Program : **Diploma in Polymer Technology**	
Course Code : **3099**	Course Title: **Rubber Technology Workshop**
Semester : **3**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0,P:3)**	Periods per semester:**45**

Course Objectives:

- To perform Mastication and compounding of NR.
- To perform Compounding of synthetic rubbers
- To perform Compression moulding, extrusion, calendering processes
- To prepare test specimens for physical testing of rubber vulcanisates.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Knowledge in chemistry		Applied Chemistry	1&2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Practice mastication and mixing of NR	23	Understanding
CO2	Practice compounding of synthetic rubbers	12	Understanding
CO3	Practice Compression moulding, extrusion and calendering processes.	8	Applying
	Lab Exam	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	**Practice Mastication and mixing of NR**		
M1.01	Identify the processing machineries mixing mill screw press and hydraulic press	3	Understanding
M1.02	Practice mastication of NR in a lab size mixing mill	3	Understanding
M1.03	Demonstrate blending of NR with SR	3	Understanding
M1.04	Prepare NR gum compound	3	Understanding
M1.05	Prepare NR masterbatch with carbon black	3	Understanding
M1.06	Prepare NR masterbatch with clay or whiting as filler	3	Understanding
M1.07	Prepare NR compound with black filler	3	Understanding
M1.08	Practice compounding of NR with non black filler	2	Understanding
	Lab Exam 1	1	
CO2	**Practice compounding of synthetic rubbers**		
M2.01	Prepare gum compound of synthetic rubbers.	3	Understanding
M2.02	Practice compounding of SBR	3	Understanding
M2.03	Practice compounding of NBR	3	Understanding
M2.04	Practice compounding of EPDM	3	Understanding
CO3	**Practice compression moulding, extrusion and calendaring processes**		
M3.01	Demonstratecompression moulding	3	Applying
M3.02	Demonstrate extrusion process	3	Applying
M3.03	Demonstratecalendering process	2	Applying
	Lab Exam2	1	

Text / Reference:

T/R	Book Title/Author
T1	Rubber Technology and Manufacture-C.M.Blow ,(publisher-Butterworth Hainemann Ltd,1982 Edition
R2	Rubber Technology -Maurice Morton (Publisher-Springer 1999 Edition)
R3	Rubber Technology Handbook -Werner Hofman (Publisher-Oxford University Press,1989 edition
R4	Vanderbilt Handbook -R.T. Vanderbilt,1990-R.T Vanderbuilt,1990 edition

Online Resources:

Sl.No	Website Link
1	Rubberchemtechnol,org
2	www.rubberatkins.com
3	https://www.science direct.com
4	https://www.utwente.nl